



Regenerative engineering of long bones using the small molecule forskolin

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Bone grafting procedures have become increasingly common in the United States, with approximately 500,000 cases occurring each year at a societal cost exceeding \$2.4 billion. Recombinant human bone morphogenetic proteins (rhBMPs) are therapeutic agents that have been widely used by orthopedic surgeons to stimulate bone tissue formation alone and when paired with biomaterials. However, significant limitations such as immunogenicity, high production cost, and ectopic bone growth from these therapies remain. Therefore, efforts have been made to discover and repurpose osteoinductive small-molecule therapeutics to promote bone regeneration. Previously, we have demonstrated that a single-dose treatment with the small-molecule forskolin for just 24 h induces osteogenic differentiation of rabbit bone marrow-derived stem cells *in vitro*, while mitigating adverse side effects attributed with prolonged small-molecule treatment schemes. In this study, we engineered a composite fibrin–PLGA [poly(lactide-co-glycolide)]-sintered microsphere scaffold for the localized, short-term delivery of the osteoinductive small molecule, forskolin. *In vitro* characterization studies showed that forskolin released out of the fibrin gel within the first 24 h and retained its bioactivity toward osteogenic differentiation of bone marrow-derived stem cells. The forskolin-loaded fibrin–PLGA scaffold was also able to guide bone formation in a 3-mo rabbit radial critical-sized defect model comparable to recombinant human bone morphogenetic protein-2 (rhBMP-2) treatment, as demonstrated through histological and mechanical evaluation, with minimal systemic off-target side effects. Together, these results demonstrate the successful application of an innovative small-molecule treatment approach within long bone critical-sized defects.

small molecules | bone regeneration | drug delivery | cyclic AMP | forskolin

Reconstruction of long bone fractures remains a major challenge in orthopedic surgery (1–3). Even with standard treatment, bone fracture patients often do not regain normal function of their affected extremity, highlighting the need for safe and clinically effective strategies for this complex clinical problem (4). Bone grafts are commonly used by surgeons to treat patients with critical-sized bone defects left by traumatic injuries or degenerative ailments (5). The current gold standard treatment options in bone grafting procedures are autografts and allografts (6). However, the use of these tissues is often associated with several limitations. For instance, the use of autografts involves donor site morbidity, limited quantities of bone for harvesting, as well as inappropriate forms (7). Allografts also present limitations such as the risk of disease transmission and immunogenicity (8).

Regenerative engineering has emerged as an innovative approach toward the formation of complex tissue, through the convergence of advanced material science, stem cell science, physics, developmental biology, and clinical transition (9–11). Within this paradigm, biodegradable polymers can be combined with adult stem cells and developmental biology cues as a viable treatment option for bone tissue regeneration and repair (12, 13). Of these development biology cues; recombinant human bone morphogenetic proteins (rhBMPs) are the most widely used agent in bone repair (14). Commercial products such as INFUSE[®] (Medtronic) have paired rhBMP-2 with absorbable collagen sponges for clinical applications such as spinal fusion and open tibial fractures (15). However, disadvantages of rhBMPs such as immunogenicity, ectopic bone growth, and high manufacturing cost prevent their widespread usage (16).

As an alternative to rhBMPs, researchers have investigated the usage of osteoinductive small molecules (17). Osteoinductive small molecules are low molecular weight compounds (<1,000 Da), that can diffuse across the cellular membrane to activate signaling cascade targets toward osteogenic differentiation (18). In comparison to protein-based growth factor treatments such as rhBMPs, these small molecules are more stable, easier to manufacture, lower in cost, and nonimmunogenic and can be delivered in a variety of formulations (19). Cyclic adenosine 3',5'-monophosphate (cAMP) is small molecule

Significance

Recombinant human bone morphogenetic proteins (rhBMPs) are extensively used agents in clinical bone repair. While constructs employing these growth factors have shown promise, disadvantages such as ectopic bone formation, and improper adipogenesis have hindered their clinical viability. In this paper, we describe a fibrin–PLGA sintered microsphere scaffold for the localized delivery of the small molecule, forskolin. We show that the release of forskolin from the scaffold promoted osteogenic differentiation of bone marrow-derived stem cells (BMSCs). When implanted in a rabbit radial critical-sized defect, the burst release of forskolin enhanced bone regeneration comparable to the matrices loaded with rhBMP-2, with minimal systemic off-target effects. This report serves as an innovative approach toward harnessing small molecules for bone regeneration.

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found ubiquitously in mammalian cells that acts as a common second messenger in controlling diverse cellular processes such as cell adhesion, cell cycle control, and cell differentiation (20). Studies have shown the capability of various cAMP analogs and cAMP activating small molecules in promoting osteogenic differentiation and mineralization through the protein kinase A (PKA)/cAMP response element-binding protein signaling cascade (21, 22). However, due to the ubiquitous nature of cAMP signaling and the small size of the cAMP-signaling small molecules, non-specific off-target effects may occur (23).

To reduce the off-target effects of these small molecules, the approach of minimizing the dosage and frequency of administration of the compounds through a short-term treatment scheme has been investigated (24, 25). This strategy allows for the increase in patient compliance, reduction in cost to the health care system, and minimization of the associated environmental impact of pharmaceutical pollutants, without compromising their therapeutic effect (26). This short-term administration approach has been shown in literature to provide long-lasting osteogenic effects in vitro and in vivo (25, 27–30). Previously, we have shown the capability of a single-dose or short-term treatment scheme (<24 h) of the cAMP-activating small molecule, forskolin, in inducing in vitro osteogenic differentiation of rabbit adipose-derived stem cells (rADSCs) and rabbit bone marrow-derived stem cells (rBMSCs), while minimizing the associated cytotoxic and anti-proliferative effects from continuous treatment schemes (31). Yet, the validation of this treatment approach within a biomaterial drug delivery scaffold for critical-sized defects and assessment toward bone regeneration in vivo has yet to be evaluated.

A number of biodegradable polymers have been paired with small-molecule compounds to repair bone tissue within a defect site (32–36). A commonly used biomaterial in bone regenerative engineering applications is poly(lactide-co-glycolide) (PLGA). PLGA is a Food and Drug Administration (FDA)-cleared, biocompatible, and biodegradable synthetic polymer that hydrolyzes into lactic and glycolic acid byproducts, which can be readily metabolized by the body (12). Our laboratory has developed osteoconductive, PLGA-sintered microsphere scaffolds that possess similar porosity and mechanical properties to native bone tissue (37, 38). Hydrogels such as fibrin can be incorporated within the pore volume of these scaffolds to facilitate the delivery of small-molecule therapeutics (39–42).

In this study, we fabricated and evaluated a composite fibrin–PLGA-sintered microsphere scaffold for the short-term delivery of forskolin. We hypothesized that by adding forskolin to the fibrin gel within the PLGA-sintered microsphere scaffold, the desired localized, burst release from the matrix will occur. When seeded with rBMSCs, results of our study showed the short-term release of forskolin from the scaffold resulted in enhanced osteogenic differentiation in vitro, without any noticeable cytotoxic effects. When implanted in a rabbit radial critical-sized defect model, the scaffold loaded with 0.8 mg of forskolin promoted bone regeneration in vivo comparable to rhBMP-2, while mitigating systemic off-target effects caused by cAMP signaling. This study reports the local, short-term delivery of forskolin from a biomaterial scaffold for long bone regeneration applications.

Results

Characterization of Forskolin-Loaded Fibrin–PLGA-Sintered Microsphere Scaffolds. The morphology of the PLGA-sintered microsphere scaffolds after the addition of fibrin gel containing

DMSO, forskolin, or rhBMP-2 was observed using SEM (scanning electron microscopy) (Fig. 1*B*). The SEM micrographs of the scaffolds at 200× magnification showed the presence of the fibrin gel within the interconnected pores formed by the adjacent microspheres. There was no significant difference in the morphology of the fibrin gel after the addition of forskolin and rhBMP-2 compared to the dimethyl sulfoxide (DMSO) control. Therefore, fibrin gel loaded with each drug component was successfully incorporated within the PLGA scaffold structure.

To confirm the short-term release of forskolin from the fibrin–PLGA-sintered microsphere scaffolds, in vitro dissolution testing in 1× phosphate-buffered saline (PBS) at 37 °C was performed (Fig. 1*C*). Within 30 min and 1 h after the onset of testing, an average of 55% and 69% of the 41.05 µg of forskolin initially loaded into the scaffold was released, respectively. At 24 h, approximately 83% of forskolin was released from the matrix, with no noticeable amount of forskolin detected at the later time points. These observations indicated that the desired rapid burst release of forskolin from the scaffold occurred at a loading efficiency of $82.6 \pm 5\%$ within the fibrin gel.

In Vitro Evaluation of Forskolin-Loaded Fibrin–PLGA-Sintered Microsphere Scaffolds. To assess the cytotoxic effects of the released forskolin and rhBMP-2 from the fibrin–PLGA-sintered microsphere scaffold, we conducted cell viability and proliferation assays on the seeded rBMSCs at days 1, 3, 7, and 14 (Fig. 2*A* and *B*). LIVE/DEAD fluorescence imaging of the seeded rBMSCs showed low cytotoxicity after treatment with each drug treatment as no obvious red signal (dead cell) difference was observed compared to the DMSO-loaded scaffolds (Fig. 2*A*). Cellular proliferation of the treated rBMSCs was evaluated quantitatively through Quant-iT PicoGreen dsDNA assay for total DNA content as an indicator for cell count (Fig. 2*B*). Compared to the DMSO drug solvent vehicle control, the proliferation of the rBMSCs amongst each treatment group showed no significant difference. Taken together, the forskolin and rhBMP-2 released from the scaffold did not induce cytotoxicity of the cells in vitro.

To evaluate the early stages of in vitro osteoblastic differentiation, the alkaline phosphatase (ALP) activity of the rBMSCs cultured on the fibrin–PLGA-sintered microsphere scaffolds loaded with DMSO, forskolin, and rhBMP-2 was measured at day 7 (Fig. 2*C*). The ALP activity of cell-seeded scaffolds in growth media was evaluated by the fold change. Compared to the scaffolds loaded with DMSO, there was a statistically significant increase in the ALP activity of the cells cultured on the forskolin and rhBMP-2-loaded scaffolds. This result shows that the forskolin and rhBMP-2 released for the fibrin–PLGA matrices were able to induce early-stage osteoblastic differentiation of the rBMSCs, similar to a direct treatment drug treatment scheme.

Quantification of the matrix mineralization by Alizarin Red staining was used as a late-stage marker of in vitro osteoblastic differentiation. The mineralization of rBMSCs on fibrin–PLGA-sintered microsphere scaffolds was quantified after 21 d of culture (Fig. 2*D*). Cells cultured on the scaffolds loaded with forskolin and rhBMP-2 showed a statistically significant increase in calcium deposition compared to the DMSO control. This shows the released forskolin and rhBMP-2 can promote matrix mineralization of rBMSCs when cultured in an osteogenic media.

In Vivo Investigation of Bone Regeneration in Rabbit Radial Bone Defect. After the implantation of the scaffolds within the radial defect, each rabbit was able to regain movement, demonstrate normal behavior, and no postoperative complications

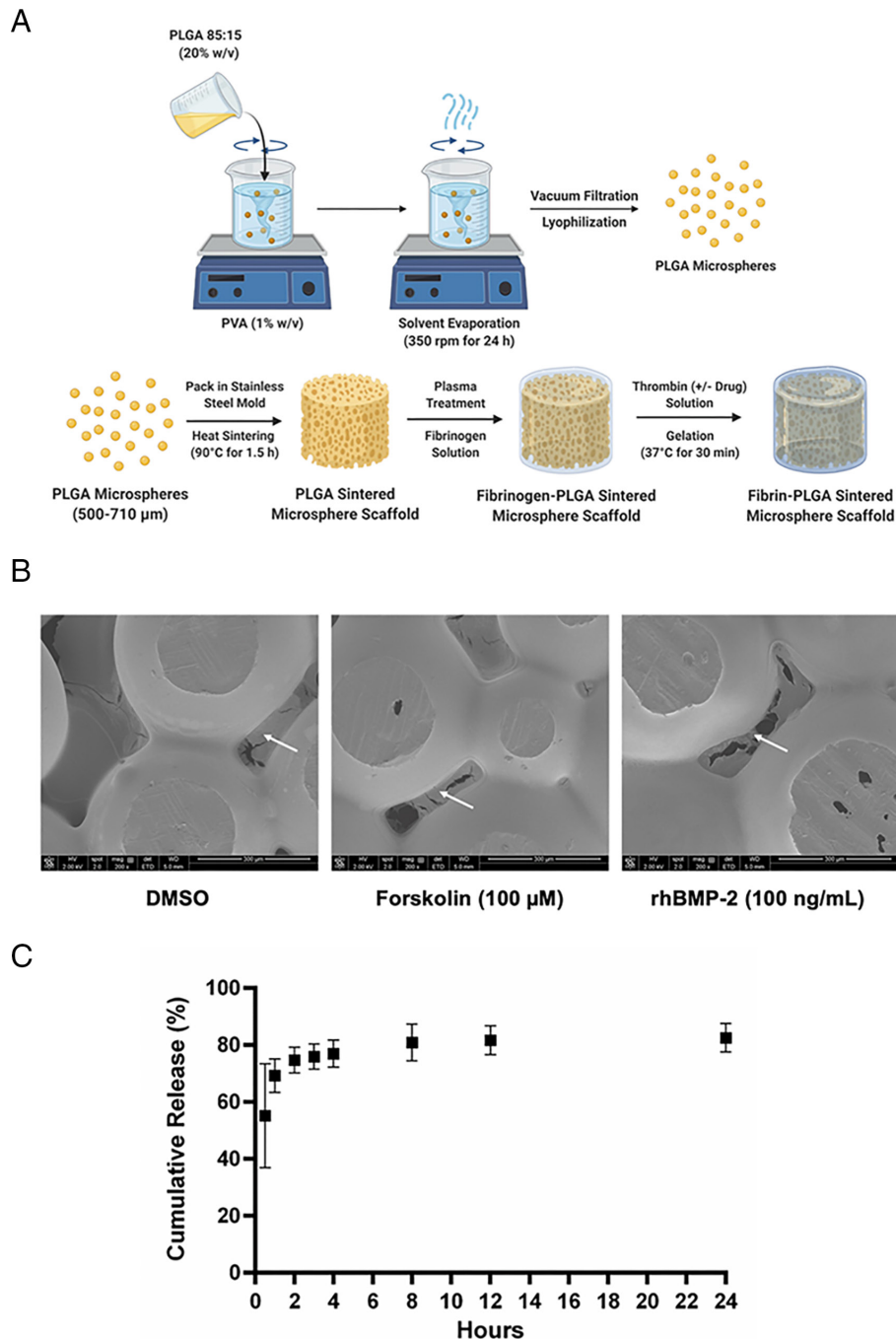


Fig. 1. Fabrication and characterization of fibrin-PLGA-sintered microsphere scaffold. (A) Schematic of fibrin-PLGA-sintered microsphere scaffold fabrication process. Created with BioRender.com. (B) Scanning electron micrographs of fibrin-PLGA microsphere scaffolds after addition of DMSO, forskolin (100 μ M), and rhBMP-2 (100 ng/mL). SEM micrographs of the scaffolds show the presence of the fibrin gel within the interconnected pores formed by the adjacent microspheres. 200 $\times$ magnification with scale bar of 300 μ m. White arrows indicate the presence of fibrin gel within pores of PLGA-sintered microsphere scaffold. (C) In vitro release kinetics of forskolin in 1 $\times$ PBS at 37 $^{\circ}$ C from fibrin-PLGA-sintered microsphere scaffolds. A rapid, burst release of forskolin from the scaffold occurs with a loading efficiency of $82.6 \pm 5\%$ within the fibrin gel. The values are presented as mean $\pm$ SD ($n = 3$).

were observed. At 12 wk postimplantation, the right forelimbs of the rabbits were harvested. No notable local reaction of the collected tissues (redness, swelling, necrosis) was detected between the experimental groups. To monitor the rate of bone healing throughout the study, X-ray imaging was conducted at weeks 0, 2, 4, 8, and 12 (Fig. 3). Over the duration of the study, a progressive increase in radiodensity within the radial defect was found in all the scaffolds, corresponding to new mineralized bone formation. At week 12, a minor increase in radiodensity was observed in both the distal and proximal ends of the defect, as well as the adjacent ulna, in the scaffolds loaded with DMSO and low dosage

of forskolin. However, with the scaffolds loaded with the high dosage of forskolin, there was a substantial increase in radiodensity in the distal end of the scaffold and the area along the ulna. In the rhBMP-2-loaded scaffolds, there was also an exuberant increase in radiodensity that filled the gap in the middle of the defect, as well as along the proximal and distal ends.

Micro-CT (Micro-Computed Tomography) analysis was conducted to evaluate the newly formed bone tissue within the radial defect at 12 wk postimplantation. Three-dimensional reconstruction and longitudinal cross-section images of the harvested forelimbs with the DMSO-loaded scaffolds displayed limited mineralized

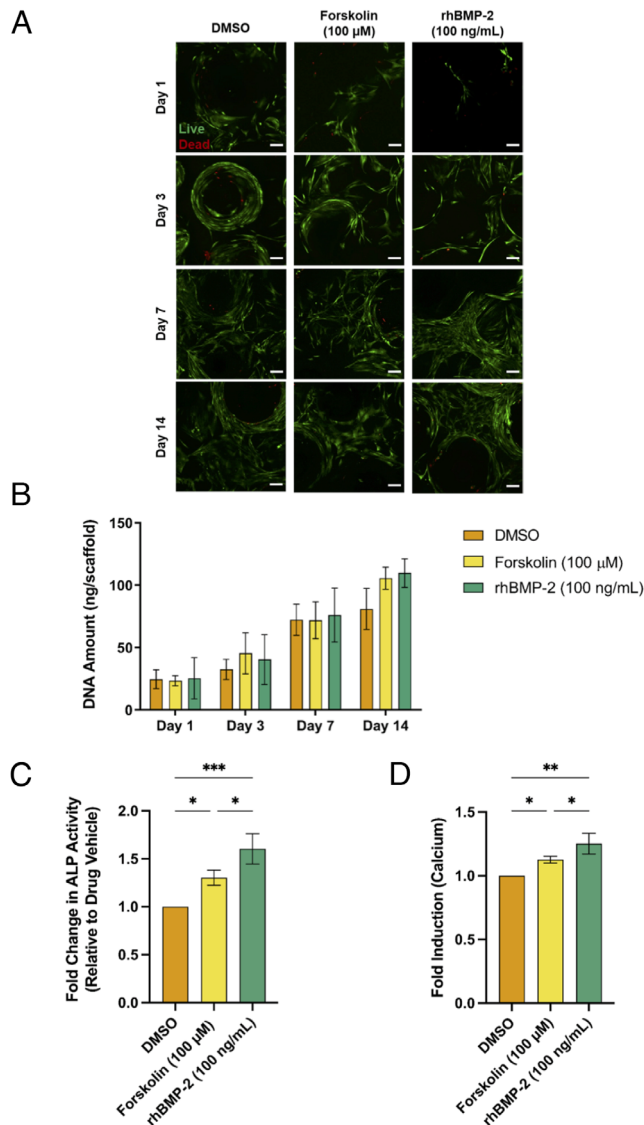


Fig. 2. In vitro evaluation of rBMSCs on fibrin-PLGA-sintered microsphere scaffolds after the addition of DMSO, forskolin (100 μM), and rhBMP-2 (100 ng/mL). (A) Representative fluorescent images of rBMSCs stained with LIVE/DEAD assay (green, calcein AM; red, ethidium homodimer-1) at days 1, 3, 7, and 14. (magnification = 10×, scale bar, 100 μm.) No obvious difference in red signal is observed in the rBMSCs seeded on the forskolin- and rhBMP-2-loaded scaffolds compared to the DMSO-loaded control scaffolds. (B) Picogreen assay of rBMSCs at days 1, 3, 7, and 14 (n = 3). No significant difference is found in the proliferation of the rBMSCs amongst each treatment group compared to the DMSO control scaffolds. Bars are sample means, and error bars denote SD. (C) Fold change in ALP activity of rBMSCs at day 7 relative to DMSO control drug vehicle (n = 3). A statistically significant increase in the ALP activity of the rBMSCs cultured on the forskolin- and rhBMP-2-loaded scaffolds is observed compared to the DMSO control. ALP activities were normalized to total DNA content. (D) Quantification of calcium deposition in rBMSCs through alizarin red S staining at day 21. rBMSCs cultured on the scaffolds loaded with forskolin and rhBMP-2 show a statistically significant increase in calcium deposition compared to the DMSO control. The fold induction of calcium was normalized relative to the DMSO control drug vehicle cultured in osteogenic medium. Bars are sample means, and the error bars denote SD (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$).

bone formation at the proximal end of radial defect (Fig. 4A). The low forskolin-loaded scaffolds demonstrated similar new bone formation along the proximal end of the defect, as well as along the adjacent ulna. In contrast, the high forskolin-loaded scaffolds showed an exuberant amount of mineralized bone formation along the distal end of the defect and throughout the area adjacent to the ulna. The rhBMP-2 scaffolds displayed almost complete

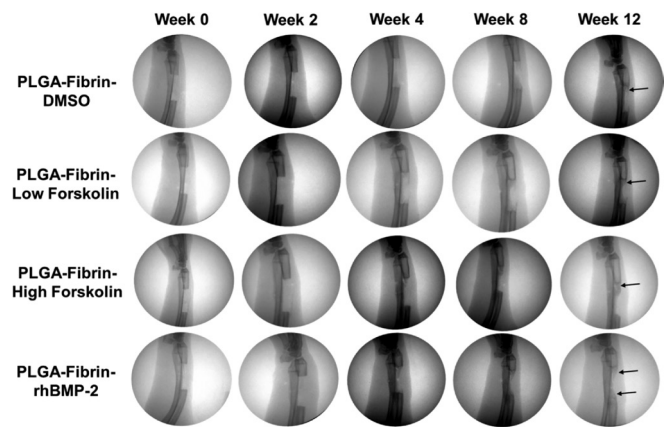


Fig. 3. Representative X-ray images of the radial bone defect at 0, 2, 4, 8, and 12 wk postimplantation. Over the duration of the study, a progressive increase in radiodensity within the radial defect is found, corresponding to new mineralized bone formation. Black arrows indicate new bone formation within the radial defect.

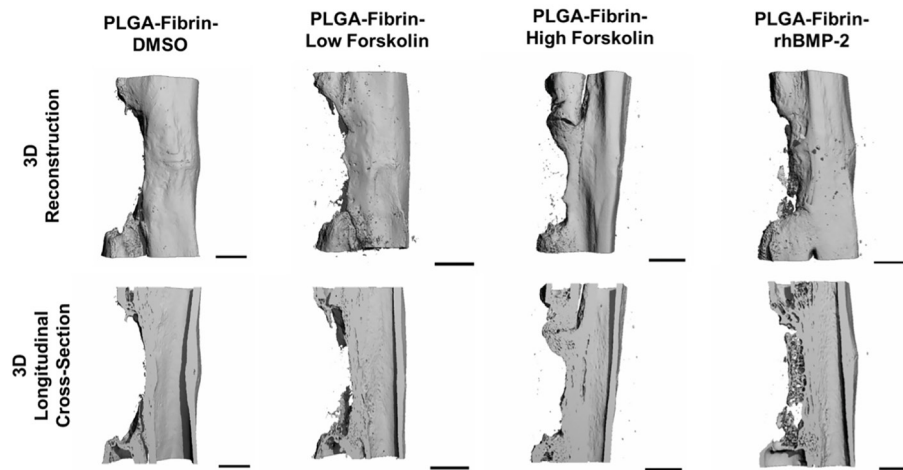
bridging of the radial defect, with new bone formation within the interior of the matrix. The induced bone volume from the micro-CT scans was quantified as Fig. 4B. At 12 wk, there was a significant increase in both bone volume and induced bone volume in both high forskolin-loaded and rhBMP-2-loaded scaffolds compared to the DMSO control. There was no statistically significant difference between the high-forskolin and rhBMP-2 scaffolds, suggesting amounts of new bone volume formed between the experimental groups were comparable.

To assess the cellular details of the harvested radius-ulna, hematoxylin and eosin (H&E) staining of the histological sections was performed (Fig. 5A). Within bone regeneration, the presence of osteoblasts and osteoclasts is an important indicator of active bone remodeling (43, 44). In the H&E-stained sections from the DMSO and low-forskolin scaffolds, minimal osteoblast and osteoclast presence was observed near the areas of new bone formation. However, an abundant number of osteoblasts (N.Ob) and osteoclasts (N.Oc) were detected on the surface of the newly formed bone in the high-forskolin and rhBMP-2 scaffolds. In addition to the higher cellular presence, blood vessels within the new bone formed at the distal end of the high-forskolin scaffolds were also observed. This indicates that vasculature was present within these matrices, which play an important role in providing nutrients to the surrounding cells.

Histological sections of the rabbit forelimbs were also stained with von kossa (VK) for mineralized bone tissue formation (Fig. 5B). The black stain represents the calcium deposits within the mineralized bone, while the pink stain represents the cytoplasm of the tissue. In the DMSO and low forskolin-loaded scaffolds, there was minimal amount of new mineralized bone tissue formed along the adjacent ulna of the defect. With the scaffolds loaded with the high dosage of forskolin, a significant amount of mineralized bone formation was found at the distal end and adjacent ulna of the radial defect. The rhBMP-2-loaded scaffolds had similar mineralized bone formation near the surface of ulna, as well as within the interior of the matrix, where the rhBMP-2 was retained within the fibrin gel.

Similar trends in new bone formation were observed in the rabbit radius-ulna sections that were stained with Goldner's trichrome (GT) (Fig. 5C). In GT staining, the green dye represents type I collagen or new bone formation, while the red dye stains for connective tissue. In addition to the new bone formation detected with the defects, a higher presence of osteoid or

A



B

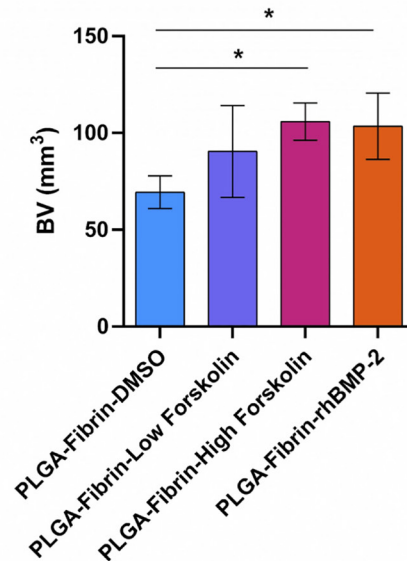


Fig. 4. MicroCT analysis of radial bone defect at 12 wk postimplantation. (A) Representative three-dimensional reconstruction and longitudinal cross-section images for each respective treatment group. The high forskolin-loaded scaffolds show an exuberant amount of bone formation along the distal end of the defect and throughout the area adjacent to the ulna. The rhBMP-2 scaffolds display a lot of bone formation in the radial defect, with new bone formation within the interior of the matrix. (Scale bar, 3 mm.) (B) Bone volume (mm^3) within the radial defect, measured from microCT. A significant increase of bone volume within the high forskolin-loaded and rhBMP-2-loaded scaffolds is detected compared to the DMSO control group. No statistically significant difference between the high-forskolin and rhBMP-2 scaffolds is also observed, suggesting comparable induced bone formation. Bars are sample means, and error bars denote SD ($n = 4$) (* $P < 0.05$).

un-mineralized bone tissue was observed in the high-forskolin and rhBMP-2 scaffolds. Also, within the distal ends of the high-forskolin scaffolds, cartilage formation was seen, which is indicative that endochondral ossification may be taking place at the defect site.

To quantify the difference in bone volume percentage (BV/TV), N.Ob per tissue area (N.Ob/T.Ar), and N.Oc per tissue area (N.Oc/T.Ar) amongst the scaffolds, static histomorphometric analysis was conducted (Fig. 6). Results of the histomorphometric analysis revealed that the high forskolin-loaded and rhBMP-2-loaded fibrin-PLGA-sintered microsphere scaffolds had a statistically significant increase of approximately 25% in BV/TV compared to the DMSO control (Fig. 6A). The N.Ob and N.Oc per tissue area in the rhBMP-2 and high-forskolin scaffolds was also significantly increased compared to the

DMSO-loaded scaffolds, thus indicating bone formation and resorption is taking place within the implant site (Fig. 6 B and C). In addition, there was no significant difference between the high-forskolin and rhBMP-2 groups in the histomorphometric parameters quantified, thus showing comparable amounts of active bone remodeling occurring.

The functionality of the newly formed bone within the scaffold implants was assessed through three-point bend testing compared to native, contralateral bone (Fig. 7). After the bending tests were completed, the max load (gf), yield load (gf), and stiffness (gf/mm) of the harvested rabbit forelimbs were calculated from the generated load vs. displacement curves. Compared to the contralateral bone, there was a significant difference in both yield load and stiffness of the DMSO- and low forskolin-loaded scaffolds (Fig. 7

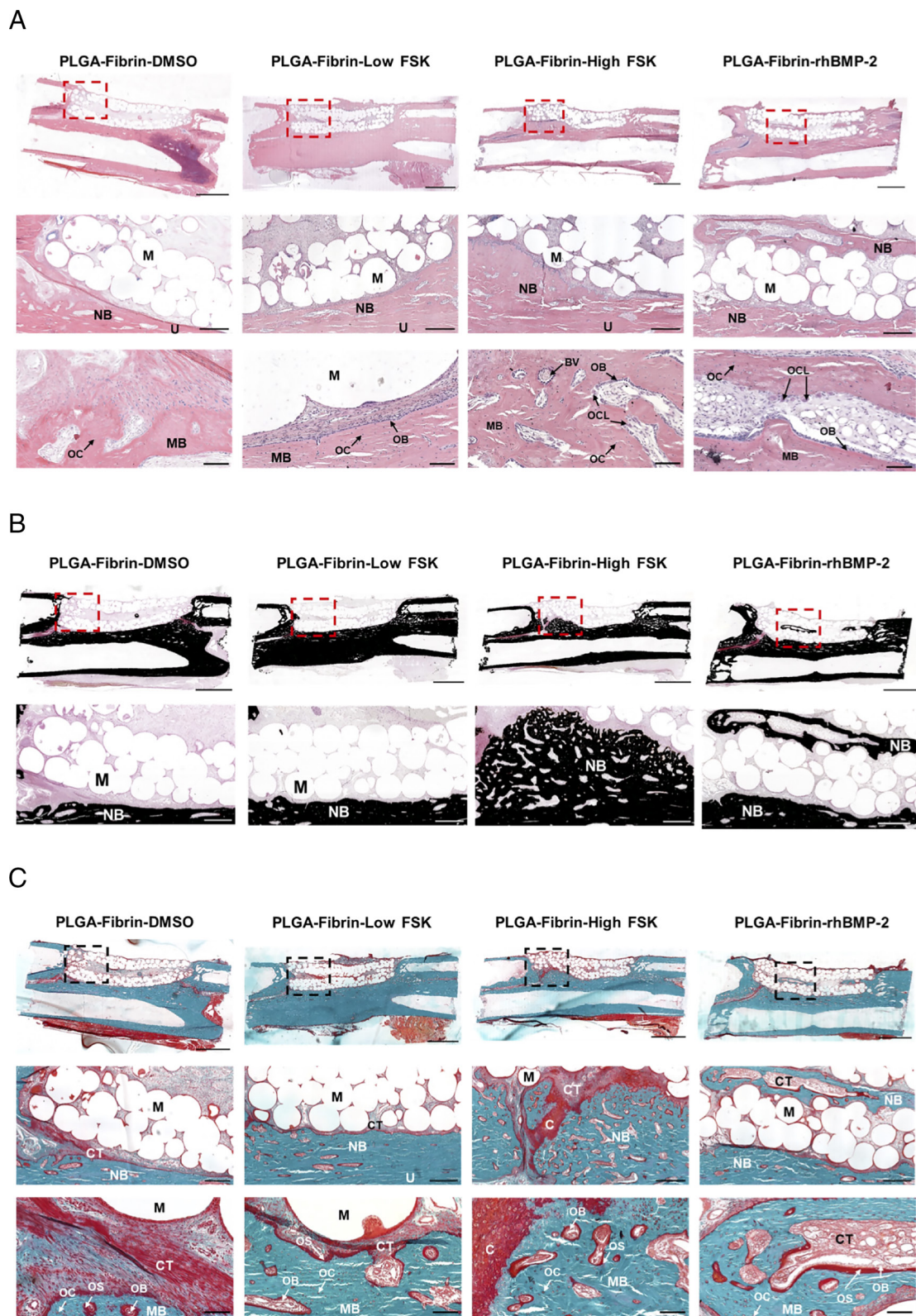


Fig. 5. Representative histological stained sections of the rabbit radius-ulna at 12 wk postimplantation. (A) Hematoxylin and eosin (H&E) staining. Osteoblasts and osteoclasts are detected in the high-forskolin and rhBMP-2 scaffolds. Blood vessels within the new bone formed at the distal end of the high-forskolin scaffolds are also observed. (B) Von Kossa staining. Black stain represents mineralized bone tissue, and the pink stain represents the cytoplasm of the tissue. Mineralized bone is found at the distal end and adjacent ulna of the radial defect in the high forskolin-loaded scaffolds. The rhBMP-2-loaded scaffolds have similar mineralized bone formation near the surface of ulna, as well as within the interior of the matrix. (C) Goldner's trichrome staining. Green dye represents new bone formation, and the red dye stains for connective tissue. Osteoid or unmineralized bone tissue is observed in the high-forskolin and rhBMP-2 scaffolds. Cartilage formation, an indicator of endochondral ossification, is observed within the distal ends of the high-forskolin scaffolds. *Top* row represents an overview of the entire defect, the *Middle* row represents the defect at 4 $\times$ magnification, and the *Bottom* row represents the defect at 20 $\times$ magnification. (Scale bar, 4 mm, 600 μ m, and 100 μ m for each row respectively.) M = microsphere, NB = new bone, U = ulna, CT = connective tissue, MB = mineralized bone, OB = osteoblast, OCL = osteoclast, OC = osteocyte, OS = osteoid, C = cartilage, BV = blood vessel.

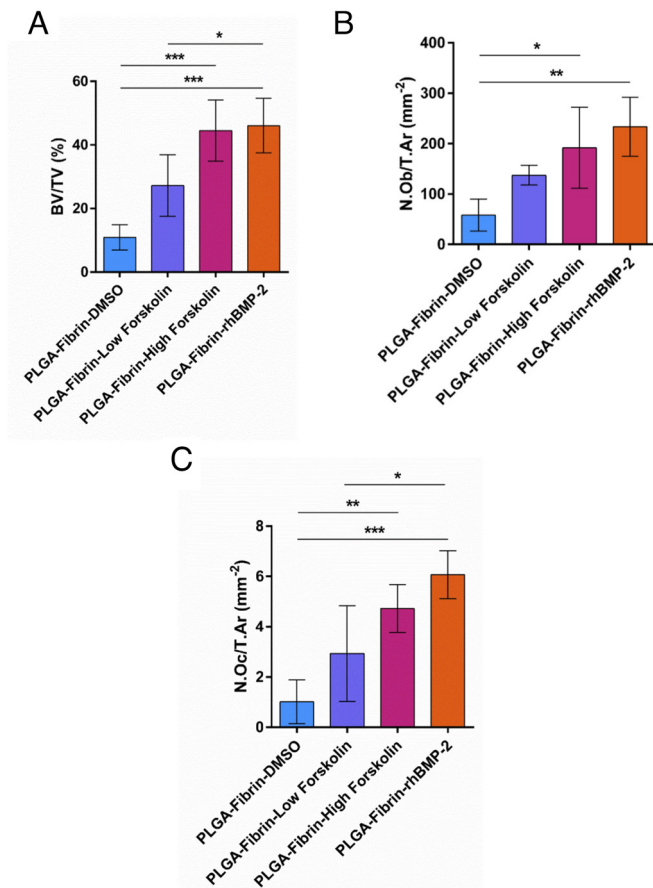


Fig. 6. Histomorphometric analysis of the harvested rabbit forelimbs at 3 mo postimplantation. (A) Bone volume/tissue volume (BV/TV). (B) Number of osteoblasts/tissue area (N.Ob/T.Ar). (C) Number of osteoclasts/tissue area (N.Oc/T.Ar). The high forskolin-loaded and rhBMP-2-loaded fibrin-PLGA-sintered microsphere scaffolds have a statistically significant increase in bone volume percentage compared to the DMSO control. The number of osteoblasts and osteoclasts per tissue area in the rhBMP-2 and high-forskolin scaffolds are also significantly increased compared to the DMSO-loaded scaffolds. Bars are sample means, and error bars denote SD ($n = 5$) (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$).

B and C). There was a lack of statistical difference in the high forskolin-loaded and rhBMP-2-loaded scaffolds, thus suggesting that the bone tissue formed within those matrices was comparable in biomechanical properties compared to native bone tissue.

In Vivo Assessment of Systemic Off-Target Effects from Forskolin-Loaded Fibrin-PLGA-Sintered Microsphere Scaffolds.

To determine whether the forskolin released from the fibrin-PLGA-sintered microsphere scaffold resulted in lipolysis or fat loss commonly associated with elevated cAMP signaling, the rabbits were weighed at various time points throughout the study. The percentage body weight change amongst each rabbit was calculated (SI Appendix, Fig. S6). Compared to the DMSO and rhBMP-2-loaded scaffolds, no significant change in body weight of the rabbits receiving forskolin treatment was observed at each respective time point.

Histopathological examination of the major organs in the rabbit such as the heart, lung, liver, kidney, and spleen was also conducted to observe potential systemic off-target effects of forskolin in vivo. In the representative H&E sections of the organs, no obvious tissue damage, pathological damage, or structural changes were observed in the forskolin-treated rabbits compared to the DMSO and rhBMP-2 experimental groups (SI Appendix, Fig. S7). Ordinal scoring of the percentage change in severity of the

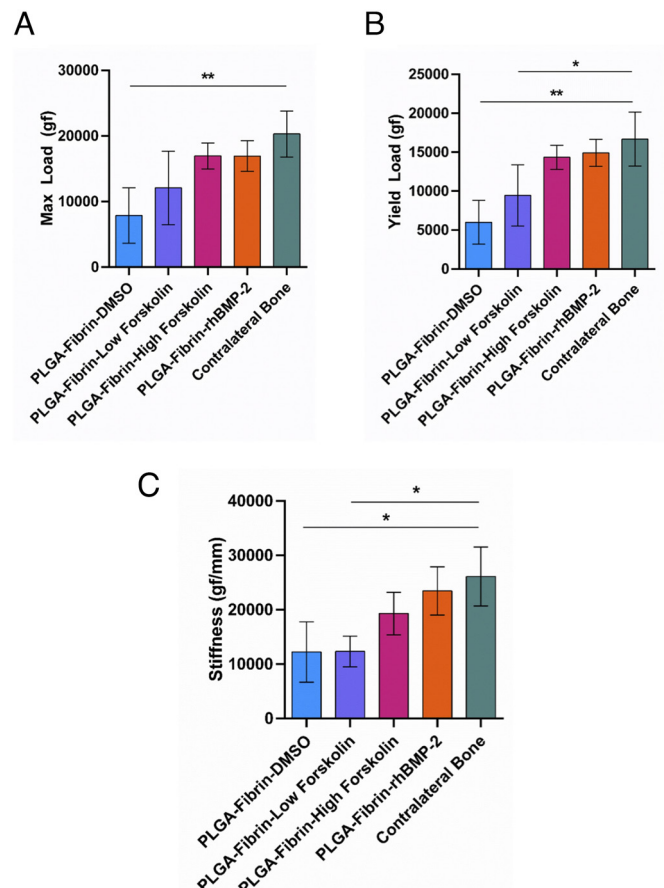


Fig. 7. Biomechanical testing of the harvested rabbit forelimbs using 3 point bending at 3 mo postimplantation. (A) Max load. (B) Yield Load. (C) Stiffness. There is no statistical difference in the yield load and stiffness of the forelimbs containing the high forskolin-loaded and rhBMP-2-loaded scaffolds compared to contralateral bone. Therefore, suggesting that the bone tissue formed within those matrices is comparable in biomechanical properties compared to native bone tissue. Bars are sample means, and error bars denote SD ($n = 3$); one-way ANOVA with Dunnett's post hoc test with contralateral bone as the control (* $P < 0.05$, ** $P < 0.01$).

examined organs was within the range of no change to <25%, thus indicating that minimal difference in organ pathology was detected (Table 1).

Comprehensive metabolic testing of the rabbit blood serum was also conducted to determine whether there is any systemic in vivo effect caused by the released forskolin. Metabolic panel results showed no statistically significant difference in the electrolyte, enzyme, total protein, cholesterol, glucose, blood urea nitrogen, creatinine, and iron levels of the rabbits implanted with the forskolin-loaded scaffolds compared to the DMSO and rhBMP-2 scaffolds (SI Appendix, Figs. S8–S12).

Discussion

Small molecule-based bone regenerative engineering has emerged as an alternative approach to circumvent the issues associated with protein-based growth factor treatment of bone defects such as recombinant bone morphogenetic proteins (rhBMPs) (17, 18, 45, 46). However, nonspecific off-target effects of these small molecules prevent their widespread usage (32). Therefore, limiting the treatment duration and frequency of these molecules present a potential approach in circumventing these off-target effects (23). In the current study, we designed and evaluated a fibrin-PLGA-sintered microsphere scaffold for the short-term delivery of forskolin. Our findings

Table 1. Histopathological analysis scores for harvested rabbit organs

	Heart			Lung		Liver			Kidney (cyst and inflammation)	Spleen neutrophilic inflammation
	Myocarditis	Necrosis/ Ischemic change	Fibrosis	Neutrophilic inflammation	Bronchitis/ Bronchiolitis	Steatosis	Inflammation	Fibrosis		
PLGA–fibrin– DMSO	0	0	0	0.2 ± 0.5	0.2 ± 0.4	0.2 ± 0.4	0	0	0	0.4 ± 0.5
PLGA–fibrin– low forskolin	0	0	0	0.4 ± 0.5	1.6 ± 1.1	0	0	0	0	0.2 ± 0.4
PLGA–fibrin– high forskolin	0	0	0	0.6 ± 1.3	0.6 ± 1.3	0.2 ± 0.4	0	0	0	0.2 ± 0.4
PLGA–fibrin– rhBMP-2	0	0	0	0.2 ± 0.4	0.6 ± 0.9	0	0.6 ± 0.5	0	0	0

Minimal difference in organ pathological scoring is detected in rabbits implanted with the forskolin-loaded scaffolds compared to the DMSO and rhBMP-2 groups. Data are represented as the mean ± SD (n = 5).

demonstrated that the burst release of forskolin from the fibrin gel incorporated within the PLGA-sintered microsphere scaffold was able to promote in vitro osteogenic differentiation of rBMSCs seeded onto the construct, without any observable cytotoxic effects. In addition, the released forskolin at an amount of 0.8 mg was able to enhance bone regeneration in vivo, while mitigating any apparent off-target effects caused by elevated cAMP signaling. These results suggest that the short-term treatment of forskolin released from a drug delivery scaffold can be utilized as an alternative treatment approach to rhBMP-2-loaded bone grafts.

A common strategy in bone regenerative engineering is to deliver bioactive factors to stimulate the development of engineered tissue (45). These factors can be incorporated within biomaterial-based scaffold systems to enhance the proliferation and differentiation of stem cells and progenitor cells (47). Fibrin is a natural biomaterial that has been used in bone regeneration applications as a local delivery system of osteoinductive bioactive factors (48, 49). During the formulation process, exogenous factors can be directly added to the fibrinogen or thrombin precursor solutions to allow uniform distribution throughout the matrix upon gelation (50). Within this loading scheme, the mesh size of the fibrin hydrogel determines the mechanism in which the drugs diffuse out of the polymer network (51). If the drug size is significantly less than the mesh size of the fibrin gel, then a fast diffusion mechanism will occur, as indicated in the Stokes–Einstein equation (52). Due to the size of forskolin of approximately 1 nm in diameter and the average mesh size of fibrin gel within the range of 100 nm in 20 mg/mL gels, a burst, short-term release of forskolin from the fibrin–PLGA-sintered microsphere scaffold took place (53).

Forskolin is a labdane diterpene compound that promotes endogenous cAMP production through the activation of the catalytic enzyme, adenylyl cyclase (AC) (54). Previously, we identified that the single-dose, short-term treatment of forskolin was capable of inducing osteogenic differentiation of rADSCs and rBMSCs in vitro. In order to translate this treatment approach into a potential critical-sized bone defect application, forskolin was added to the fibrin gel component of the engineered PLGA-sintered microsphere scaffold. 0.4 mg and 0.8 mg of forskolin were selected as the low and high dosages in the in vivo bone regeneration studies because they represent 10× and 20× of the amount used in vitro, as well as being within the range of 0.01 to 0.1 mg/kg of forskolin used in previous localized treatments (55). Results of the in vivo bone regeneration studies showed that the high-forskolin dosage of 0.8 mg was capable of promoting new bone formation at the distal end of the radial defect and along the adjacent ulna. This observation can be due to the contact of the released forskolin with the bone marrow stromal cells localized within the distal end

of the defect and the osteoprogenitor cells within the periosteum of the ulna.

Evidence of endochondral ossification through cartilage formation within the high forskolin-loaded scaffolds was also observed. Endochondral ossification is the predominant regenerative process in long bones, where a hyaline cartilage intermediate is replaced with mineralized bone tissue (56). Forskolin treatment has been previously shown in literature to promote enhanced hyaline cartilage formation in human pluripotent stem cells in vitro (57). Therefore, within a long bone microenvironment, the released forskolin from the fibrin–PLGA scaffold may have induced the differentiation of the surrounding stem cells toward a chondrogenic lineage, as depicted in the GT-stained sections, thus, resulting in enhanced hyaline cartilage intermediate production and subsequent mineralized bone tissue formation at the defect site.

Vascularization plays an important role in tissue regeneration to provide necessary nutrients and oxygen to the surrounding cells in bone tissue, while removing metabolic waste (58). To enhance the vasculature within engineered constructs, growth factors such as vascularized endothelial growth factor (VEGF) have been delivered from the materials to facilitate angiogenesis (28). H&E staining of the scaffolds containing the high-forskolin dosage showed the presence of blood vessels within the newly formed bone tissue at the distal end of the defect. Previous studies have shown the capability of forskolin to enhance angiogenesis through the production of VEGF via the coordinated cross-talk between PKA-dependent and Epac-dependent signaling in endothelial cells (59). Therefore, the induction of VEGF secretion by forskolin from the surrounding cells in the radial defect may have provided this enhanced vascularization in the forskolin-loaded scaffold implants.

cAMP signaling is a ubiquitous secondary messenger system that regulates a variety of physiological effects in mammalian cells (60). It is known that the activation of AC and accompanying increase in intracellular cAMP levels from forskolin treatments can promote undesired off-target pharmacological effects outside of the bone microenvironment (61). For example, elevated cAMP signaling from forskolin administration has been shown to decrease blood pressure and systemic vascular resistance, decrease steatosis in the liver, and promote activation of hormone-sensitive lipase to increase lipolysis in adipose tissue (62). In order to assess the capability of the short-term release of forskolin from the fibrin–PLGA-sintered microsphere scaffolds in mitigating these off-target effects, the percentage weight change, histopathological examination of the major organs, and metabolic blood tests of the treated rabbits were conducted. Results of these tests showed no observable difference in the systemic levels of the rabbits compared to the DMSO and

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rhBMP-2 treatment groups, therefore, confirming the efficacy of the short-term treatment scheme of forskolin *in vivo*.

In conclusion, we conducted a comprehensive evaluation of the short-term delivery of forskolin from a fibrin–PLGA-sintered microsphere scaffold for bone regenerative engineering applications. The scaffolds were shown to retain the bioactivity of forskolin through promoting osteogenic differentiation of BMSCs, while mitigating cytotoxic effects *in vitro*. When implanted in a rabbit radial critical-sized defect, the released forskolin from the scaffolds also enhanced bone regeneration *in vivo*, while mitigating systemic off-target effects observed from elevated cAMP signaling. Future studies will focus on incorporating forskolin with osteoinductive materials, evaluating the effect of short-term treatment of forskolin on *in vitro* osteogenic differentiation with human mesenchymal stem cells, modulating the release kinetics of the small molecule to enhance its therapeutic window and efficacy, and assessing a synergistic treatment scheme with chondrogenic small molecules toward promoting endochondral ossification *in vivo*.

Materials and Methods

PLGA Microsphere Preparation. PLGA microspheres were prepared using an oil-in-water emulsion solvent evaporation method as previously described (63). Briefly, 4 g of PLGA (85:15) was dissolved in 20 mL of methylene chloride to create a 20% w/v solution. The solution was subsequently added dropwise into a 1% polyvinyl alcohol (PVA) solution mixed with a magnetic stirrer at 350 rotations per minute (RPM) for 24 h. The resulting microspheres were isolated by vacuum filtration, rinsed with distilled water, and lyophilized overnight for 24 h. The dried microspheres were sieved to a diameter range of 500 to 710 and stored under vacuum in a desiccator for future use.

Fibrin–PLGA-Sintered Microsphere Scaffold Fabrication. PLGA-sintered microsphere scaffolds were prepared through a heat sintering method as previously optimized in our laboratory (64). PLGA microspheres were packed into a disc or cylindrical-shaped stainless-steel mold with the dimensions of 7 mm × 2 mm or 15 mm × 5 mm, respectively (SI Appendix, Fig. S1). The mold containing the microspheres was placed in a vacuum oven and heated at 90 °C for 90 min. After the sintering time was complete, each mold was allowed to cool to room temperature before removal of the scaffolds.

The scaffolds were sterilized through immersion in 70% ethanol for 30 min, washing three times with sterile distilled water, and exposure to UV for 30 min on each side. Prior to the addition of fibrinogen, the scaffolds were O₂ plasma treated for 5 min at 0.5 Torr and 100 W (CUTE-1MP/R, Gyeonggi, Korea) to enhance their wettability. After plasma treatment, 7.5 µL or 30 µL of fibrinogen solution (20 mg/mL, 0.9% NaCl) was pipetted directed to the surface of the disc or cylindrical scaffold, respectively. Once the fibrinogen solution fully soaked into the construct, 7.5 µL or 30 µL of thrombin solution (20 U/mL, 0.1% bovine serum albumin (BSA)) was added to the scaffold and placed at 37 °C for 30 min to allow formation of the fibrin gel. For the fibrin–PLGA scaffolds containing DMSO, forskolin, or rhBMP-2, 0.7 µL or 3 µL of each component was added to the 7.5 µL or 30 µL thrombin solution and thoroughly homogenized through sonication before gelation. For the *in vitro* cell studies, 41.05 µg of forskolin and 100 ng of rhBMP-2 were used. For the *in vivo* studies, 0.4 mg and 0.8 mg of forskolin were added for the low- and high-forskolin experimental groups respectively, whereas 15 µg of rhBMP-2 was added for the positive control group (Fig. 1A).

Morphology of Fibrin–PLGA-Sintered Microsphere Scaffolds. The morphology of the fibrin–PLGA-sintered microsphere scaffolds was visualized using SEM. The scaffolds were first freeze-dried for 24 h. After lyophilization, the samples were mounted on SEM stubs using carbon tape, sputter coated with gold/Palladium (E5100, Polaron), and observed using a scanning electron microscope (Nova NanoSEM 450, FEI).

In Vitro Release Studies. The scaffolds were submerged in centrifuge tubes containing 1 mL of preheated 1 × PBS and placed at 37 °C under gentle rocking. At predetermined time points, 500 µL aliquots of the medium were withdrawn and replaced by an equal volume of fresh prewarmed medium. The amounts of

forskolin released in the collected samples were quantified through high-performance liquid chromatography using the Agilent Technology 1260 Infinity II system. The mobile phase consisted of (A) acetonitrile and (B) H₂O in a 65A/35B ratio at a flow rate of 1 mL/min. The reverse-phase column was a 5-µm Discovery® HS C18 maintained at 25 °C, and the run time was 10 min. The sample solution was injected at a volume of 10 µL, and the detection wavelength was 210 nm. A standard curve of known forskolin concentrations was generated to determine the amount of forskolin in each release sample.

Cell Isolation and Culture. rBMSCs were isolated from male New Zealand white rabbits (3 to 4 kg) in accordance with guidelines and regulations approved by University of Connecticut Health Center Institutional Animal Care and Use Committee (IACUC) protocol (TE-101976-0122). The isolation and purification of the rBMSCs were performed according to our previously optimized protocol (31). Characterization of the isolated rBMSCs was conducted using flow cytometry for CD44, CD11b, and CD45 cell-surface markers, as well as *in vitro* differentiation assays into multiple lineages (bone and adipose) (SI Appendix, Fig. S13).

For the *in vitro* cell studies, rBMSCs were seeded onto each respective scaffold at a density of 50,000 cells per scaffold. The cells were allowed to adhere for 1 h to allow for cell attachment before adding the growth media. The cell-seeded scaffolds were maintained in Dulbecco's Modified Eagle Medium (DMEM)-F12 with 10% fetal bovine serum (FBS) and 1% P/S. For mineralization studies, cells were cultured in DMEM-F12 supplemented with 10% FBS, 1% antibiotics, 3 mM β-glycerophosphate, 10 µg/mL ascorbic acid, and 100 nM of dexamethasone. The cells were incubated in a humidified incubator at 37 °C containing 5% CO₂.

Cell Viability. The viability of the cells seeded on the scaffolds at days 1, 3, 7, and 14 was assessed using the LIVE/DEAD™ Viability/Cytotoxicity Kit (Invitrogen, Carlsbad, CA) according to the manufacturer's instructions. The scaffolds were transferred into new wells, washed with 1 × PBS, and incubated in the staining solution (0.5 µL/mL calcein-AM, 4 µL/mL ethidium homodimer-1 in 1 × PBS) for 20 min at room temperature. The stained cells were rinsed twice with 1 × PBS and imaged using confocal microscopy (Zeiss LSM 880) at 10× magnification.

Cell Proliferation. The cell proliferation study was performed using the Quant-iT PicoGreen dsDNA assay kit (Invitrogen, Carlsbad, CA) following the manufacturer's instructions. At days 1, 3, 7, and 14, the scaffolds were harvested and washed with PBS before being transferred into a new 24-well plate. The scaffolds were lysed with 0.1% Triton X-100 and underwent three freeze–thaw cycles. The cell lysates were mixed with the Quant-iT PicoGreen reagent and measured via spectrophotometry at 535 nm with excitation at 485 nm, and the DNA content was determined through a standard curve.

ALP Activity. ALP activity was measured using an ALP substrate assay kit (Bio-Rad), according to the manufacturer's instructions. At day 7, the scaffolds were washed three times with PBS, lysed with 0.1% Triton X-100, and underwent three freeze–thaw cycles. The cell lysates were collected and incubated in the ALP substrate solution for 1 h at 37 °C. The color change due to the conversion of p-nitrophenyl phosphate into p-nitrophenol in the presence of ALP was analyzed by spectrophotometry at 405 nm. The ALP activities were normalized to total DNA content using a Quant-iT PicoGreen dsDNA assay kit (Invitrogen, Carlsbad, CA). ALP activities were further normalized to the activities expressed by the scaffolds containing the drug delivery vehicle (DMSO).

Matrix Mineralization. Mineralized matrix deposition was evaluated using the alizarin red staining method for calcium deposition. At day 21, the scaffolds were rinsed three times with PBS to remove unattached cells. The cells were fixed in 70% ethanol for 1 h at 4 °C and allowed to air-dry for 5 to 10 min. Following a wash with distilled, deionized water (DDI H₂O), the samples were covered in 40 mM of alizarin red solution (Sigma-Aldrich) and incubated with shaking for 20 min at room temperature. After washing the scaffolds 5 times with DDI H₂O, 10% cetylpyridinium chloride was added to solubilize the red matrix precipitate, yielding a purple solution. The optical density of the solution was read at 562 nm using a BioTek plate reader. The calcium deposition was normalized to the control DMSO samples cultured in a growth medium.

In Vivo Study Design. All animal experiments were approved by the IACUC at the University of Connecticut Health under protocol number, TE-102048-0322. Forty New Zealand White rabbits (3 to 4 kg) were used in this study and

were divided into four groups ($n = 10$, each) for 12 wk: 1) PLGA-fibrin-DMSO, 2) PLGA-fibrin-low forskolin (0.4 mg), 3) PLGA-fibrin-high forskolin (0.8 mg), and 4) PLGA-fibrin-rhBMP-2 (15 μ g). Based on power analysis ($\beta = 80\%$, $\alpha = 5\%$), four rabbits were randomly designated for histological analysis and six rabbits for biomechanical testing. Blood serum analysis ($n = 5$) and three-point bending tests ($n = 3$) were conducted based on sample availability.

Blood Collection and Serum Analysis. Whole blood from the marginal ear vein of the rabbits was collected preoperatively and on days 2, 7, 14, 28, 42, 56, 70, and 84 of the 12-wk study. Prior to blood collection, the rabbits were sedated with acepromazine (0.25 to 1 mg/kg). The blood was then drawn into BD Vacutainer[®] serum tubes (BD, Franklin Lakes, NJ) and allowed to clot for 30 min at room temperature. The samples were centrifuged at $2,000 \times g$ for 20 min at 4 °C. The resulting supernatants containing the serum were collected, transferred to Eppendorf tubes, and frozen at -80 °C. A small animal chemistry panel of the collected rabbit serum was conducted at the Animal Health Diagnostic Center, Cornell University College of Veterinary Medicine.

X-Ray Imaging. To confirm the successful creation of the radial defect and to monitor the progress of bone formation throughout the duration of the study, radiological images of the rabbit forelimbs were taken at weeks 0, 2, 4, 8, and 12 using a digital mobile C-Arm (MINI6600[™] OEC Medical Systems).

Micro-CT. Bone microarchitecture of the rabbit forelimbs was determined using a micro-CT instrument (μ CT 40, SCANCO Medical AG, Bassersdorf, Switzerland), which was calibrated weekly using a phantom provided by the manufacturer. The rabbit forelimbs were submerged in 70% ethanol and scanned at 16- μ m resolution with an energy level of 55 peak kV, intensity of 145 μ A, and integration time of 300 ms. Contours were drawn to define the region of interest and were analyzed with a lower threshold of 578.8, upper threshold of 2088.7, Gauss sigma of 0.8, and Gauss support of 1. Three-dimensional reconstruction images of the radius-ulna were generated, and the regions within the bone defect were assessed for bone volume.

Bone Histology and Histomorphometry. The rabbit forelimbs were processed for histology using methylmethacrylate embedding procedure outlined by Erben et al. (65). After fixation in 70% ethanol, the samples were rinsed overnight under running tap water, dehydrated with alcohols (95% ethanol and 100% isopropanol), and cleared with xylene. The tissues were subsequently infiltrated with the polymethyl methacrylate (PMMA) embedding medium, trimmed, and sectioned into 8- μ m-thick slices using a Reichert-Jung Ultracut E microtome equipped with a tungsten carbide blade. The sections were then mounted onto glass slides, deplasticized, and stained for H&E (Fig. 5A), VK (Fig. 5B), GT (Fig. 5C), toluidine blue (TB) (SI Appendix, Fig. S3), ALP (SI Appendix, Fig. S4), and tartrate-resistant acid phosphatase (SI Appendix, Fig. S5). Imaging was conducted under a light microscope scanner (Aperio CS2, Leica Microsystems).

Histomorphometric measurements of the sectioned rabbit forelimbs were conducted using OsteoMeasure software (Osteometrics, Atlanta, GA, USA). Forty images at 40 $\times$ magnification of the TB-stained sections were analyzed along the distal, middle, and proximal areas of the radial bone defect per rabbit sample. BV/TV, %, N.Ob/T.Ar, mm⁻², and N.Oc/T.Ar, mm⁻² were measured. All terminology

and units used are those recommended by the Histomorphometry Nomenclature Committee of the American Society for Bone and Mineral Research (66).

Biomechanical Testing. The frozen forelimbs were thawed slowly at 4 °C overnight and equilibrated to room temperature for a few hours before testing. The soft tissue surrounding the harvested radius and ulna was dissected, leaving only the bone structures in place. To exclude the influence of the adjacent ulna from the mechanical test, two transversal cuts from each side of the ulna were made at approximately 90% of its width and 10 mm apart from each other using a surgical microsaw (Stryker RemB 6400-34) (67). Three-point bending was carried out using a Mach-1 Mechanical Tester Model v500csst (Biomomentum) equipped with a 25-kg load cell. The load cell was calibrated prior to use. The radius and ulna were placed horizontally onto two supporting beams spaced at a distance of 18 mm apart within a testing chamber. The loading piece of the apparatus was moved down until it was approximately 2 mm away from the midpoint of the sample, and the test was conducted at a stage velocity of 0.1 mm/s until failure. Throughout the testing, the samples were immersed in phosphate-buffered saline maintained at body temperature (37 °C) to replicate in vivo conditions. Load vs. displacement curves were analyzed using Mach-1 software to determine the maximum load (gf), yield load (gf), and stiffness (gf/mm) of the harvested rabbit forelimbs. The contralateral forelimbs were tested as a positive control for comparison to native bone.

Statistical Analysis. GraphPad Prism 6 (GraphPad Software; San Diego, CA) was used for statistical analysis and graph design. Quantitative data were reported as mean $\pm$ SD. Statistical analysis was performed using a one-way ANOVA. Comparison between the two means was determined using the Tukey test with statistical significance evaluated at $*P < 0.05$, $**P < 0.01$; $***P < 0.001$, and $****P < 0.0001$.

Data, Materials, and Software Availability. All study data are included in the article and/or supporting information.

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